

Little Sub and Enigma

Tags:

Time Limit: 1 s Memory Limit: 256 MB
Submission: 2146 AC: 234 Score: 88.00

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Description

Little Sub builds a naive Enigma machine of his own. It can only be used to encrypt/decrypt lower-case letters by giving each letter a unique corresponding lower-case letter. In order to ensure the accuracy, no contradiction or controversy is allowed in both the decryption and the encryption, which means all lower-case letters can only be decrypted/encrypted into a distinct lower-case letter.

Now we give you a string and its encrypted version. Please calculate all existing corresponding relationship which can be observed or deduced through the given information.

Input

The first line contains a string S , indicating the original message.

The second line contains a string T , indicating the encrypted version.

The length of S and T will be the same and not exceed 1000000.

Output

we use a string like 'x->y' to indicate that letter x will be encrypted to letter y.

Please output all possible relationships in the given format in the alphabet order.

However, if there exists any contradiction in the given information, please just output Impossible in one line.

Samples

input [Copy](#)

```
banana
cdfdfd
```

output [Copy](#)

```
a->d
b->c
n->f
```

input [Copy](#)

```
banana
cdfdfs
```

output [Copy](#)

```
Impossible
```

Author

[YE, Zicheng](#)

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[HZNUOJ](#) is based on [HUSTOJ](#)

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Server Time: 2025-8-26 16:08:07